04

Sentiment Analysis

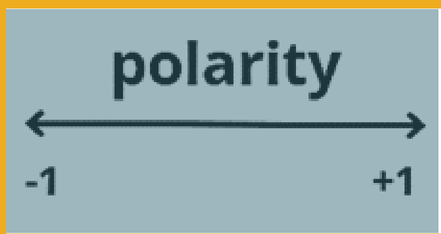
NLP Techniques

- Sentiment analysis
- Topic modeling
- Text generation

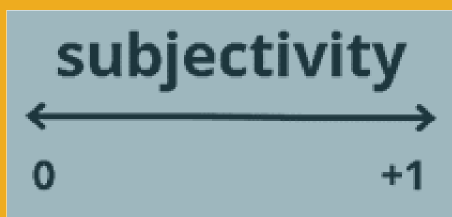
Sentiment analysis

- Input : A corpus. The reason we're not using the document-term matrix here is because order matters. "great" = positive. "not great" = negative
- TextBlob : TextBlob is a Python library build on top of NLTK. It's easier to use and provides some additional functionality, such as rule-based sentiment scores.
- Output : Each movie will be given a sentiment score and subjectivity score.

```
# The code for TextBlob's sentiment analysis
from textblob import TextBlob
Textblob("I love Indonesia").sentiment
```




```
Sentiment(polarity=0.5, subjectivity=0.6)
```



How does TextBlob work?

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```
<word form="abhorrent" wordnet_id="a-1625063" pos="JJ" sense="offensive to the mind" polarity="-0.7" subjectivity="0.8" intensity="1.0" r
<word form="able" cornetto_synset_id="n_a-534450" wordnet_id="a-01017439" pos="JJ" sense="having a strong healthy body" polarity="0.5" su
<word form="able" wordnet_id="a-00001740" pos="JJ" sense="(usually followed by 'to') having the necessary means or skill or know-how or a
<word form="able" wordnet_id="a-00306663" pos="JJ" sense="having inherent physical or mental ability or capacity" polarity="0.5" subjecti
<word form="able" wordnet_id="a-00510348" pos="JJ" sense="have the skills and qualifications to do things well" polarity="0.5" subjectivi
<word form="above" cornetto_synset_id="n_a-504850" wordnet_id="a-00125993" pos="JJ" sense="appearing earlier in the same text" polarity="
<word form="abridged" cornetto_synset_id="d_a-9176" wordnet_id="a-00004413" pos="JJ" sense="(used of texts) shortened by condensing or re
<word form="abrupt" cornetto_synset_id="n_a-505100" wordnet_id="a-00640520" pos="JJ" sense="surprisingly and unceremoniously brusque in m
<word form="abrupt" cornetto_synset_id="n_a-529169" wordnet_id="a-01145151" pos="JJ" sense="extremely steep" polarity="0.0" subjectivity=
<word form="abrupt" wordnet_id="a-01143585" pos="JJ" sense="exceedingly sudden and unexpected" polarity="0.0" subjectivity="1.0" intensit
<word form="abrupt" wordnet_id="a-02294122" pos="JJ" sense="marked by sudden changes in subject and sharp transitions" polarity="0.0" sub
```


Word Form	Sense	Polarity	Subjectivity
great	very good	1.0	1.0
great	of major significance or importance	1.0	1.0
great	relatively large in size or number or extent	0.4	0.2
great	remarkable or out of the ordinary in degree or magnitude effect	0.8	0.8

Word Form	WordNet ID	POS	Sense	Polarity	Subjectivity
great	a-01123879	JJ	very good	1.0	1.0
great	a-012378818	JJ	of major significance or importance	1.0	1.0
great	a-01386883	JJ	relatively large in size or number or extent	0.4	0.2
great	a-01677433	JJ	remarkable or out of the ordinary in degree or magnitude effect	0.8	0.8

`Textblob(“I love Indonesia”).sentiment`	`Sentiment(polarity=0.8, subjectivity=0.75)`
`Textblob(“I love Indonesia”).sentiment`	`Sentiment(polarity=0.4, subjectivity=0.75)` polarity x-0.5
`Textblob(“I love Indonesia”).sentiment`	`Sentiment(polarity=1, subjectivity=0.75)` subjectivity x 1.3
`Textblob(“I love Indonesia”).sentiment`	`Sentiment(polarity=0.8, subjectivity=0.75)`

TextBlob Sentiment

TextBlob finds all of the words and phrases that it can assign a polarity and subjectivity to, and averages all of them together

Output: each movie is assigned one polarity and one subjectivity score

While not the most sophisticated technique, it’s a good starting point

This rule-based implementation is a knowledge-based technique. There are also statistical methods out there such as Naive Bayes

Picture Source

- unsplash.com
- pexels.com
- pixabay.com